

FL Simulation Results

File: test_Median

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 3
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 100.0% (1.00)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 100.0% (1.00) of data for 15 rounds with 3 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8715
Clean + DP Protection Accuracy: 0.8532
Poisoned Accuracy: 0.4568
Poisoned + DP Protection Accuracy: 0.5475
Drop (Clean - Poisoned): 0.4146
Drop (Clean - Init): 0.0787
Drop (Clean DP - Init): 0.0604
Drop (Poisoned - Init): -0.3360
Drop (Poisoned DP - Init): -0.2453
DP Protection Method: median
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8782
Clean Recall: 0.8715
Clean F1 Score: 0.8711

Clean DP Precision: 0.8652
Clean DP Recall: 0.8532
Clean DP F1 Score: 0.8531

Poisoned Precision: 0.5878
Poisoned Recall: 0.4568
Poisoned F1 Score: 0.4725

DP Protection Precision: 0.7015
DP Protection Recall: 0.5475
DP Protection F1 Score: 0.5554

Summary

FL Simulation Complete
Task: 2b93db49-5c4e-4ad9-a588-844b5d1cb138
GPU: GPU 0
Init Accuracy: 0.7928
Clean Accuracy: 0.8715
Clean DP Accuracy: 0.8532
Poisoned Accuracy: 0.4568
Data Poison Protection Accuracy: 0.5475
Drop (Clean - Poisoned): 0.4146
Drop (Clean - Init): 0.0787
Drop (Clean DP - Init): 0.0604
Drop (Poisoned - Init): -0.3360
Drop (Poisoned_DP - Init): -0.2453
Data Poison Protection Method: median
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8782
Clean Recall: 0.8715
Clean F1 Score: 0.8711
Clean DP Precision: 0.8652
Clean DP Recall: 0.8532
Clean DP F1 Score: 0.8531
Poisoned Precision: 0.5878
Poisoned Recall: 0.4568
Poisoned F1 Score: 0.4725
DP Protection Precision: 0.7015
DP Protection Recall: 0.5475
DP Protection F1 Score: 0.5554